FIFA WORLD CUP
HOTEL NETWORK DESIGN AND IMPLEMENTATION

Cisco Packet Tracer Project Report

Student ID: 22101777

Base Network: 17.77.0.0/16

Network design, VLSM addressing, routing, DHCP, DNS, web, email, failover, and verification

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1. Executive Summary

This report documents the design and implementation of a FIFA World Cup hotel network in Cisco Packet Tracer for six country hotels: France, Spain, Germany, Argentina, Brazil, and Portugal. The design provides separate hotel LANs, inter-router connectivity, dynamic and static routing, DHCP, centralized DNS, web access, email communication, printing, and a backup path for the Portugal-France link.

The addressing plan is derived from Student ID 22101777. The last four digits, 1777, are divided into 17 and 77, producing the base network 17.77.0.0/16. Variable Length Subnet Masking (VLSM) is then used to allocate each hotel a subnet sized for its required host population while conserving address space.

Verification evidence demonstrates active router interfaces, RIP-learned routes, Brazil's default route through Argentina, successful DHCP leases, centralized DNS resolution, access to both FIFA web servers, inter-hotel email communication, end-to-end ICMP reachability, and a controlled link-failure test.

2. Project Requirements and Objectives

The project requires one router per country hotel and a topology that follows specific physical and logical relationships. Brazil connects only to Argentina. France, Spain, and Argentina share a switched transit segment. The European routers form a loop: France-Portugal-Spain-Germany-France. Each hotel also requires an email server, a printer, and two PCs to represent the larger host population.

Hotel / Country	Required Hosts
France	200
Spain	100
Germany	150
Argentina	200
Brazil	250
Portugal	300

Table 1. Required host capacity by hotel.

- Argentina and France provide dedicated web servers at www.argentina.fifa.com and www.france.fifa.com.
- Argentina hosts the centralized DNS service used by all hotels.
- France router provides DHCP for France, Spain, Germany, and Portugal; Argentina and Brazil use dedicated DHCP servers.
- France, Spain, and Germany exchange routes through RIP.
- Argentina and Portugal use static routing; Brazil reaches all other networks through a default route to Argentina.
- Portugal must retain connectivity through an alternate path when the direct Portugal-France link is unavailable.
- All end devices must be able to reach the required networks and services.

3. Design Assumptions and Methodology

Cisco 2911 routers and 2960 switches are used in the Packet Tracer implementation. Serial point-to-point links are used for the European loop and the Argentina-Brazil connection, while GigabitEthernet interfaces connect routers to local switches and to the France-Spain-Argentina transit switch. HWIC serial modules are assumed where required.

Low-numbered addresses in each LAN are reserved for the gateway, servers, and printer. Client PCs receive addresses dynamically. The central DNS server address is 17.77.4.11. Router and infrastructure addresses are configured manually. A DCE serial side uses a clock rate of 64000 where Packet Tracer requires it.

4. Addressing Plan and VLSM

4.1 Base Network Derivation

Student ID 22101777 -> last four digits 1777 -> 17 and 77 -> base network 17.77.0.0/16.

4.2 VLSM Allocation Tree

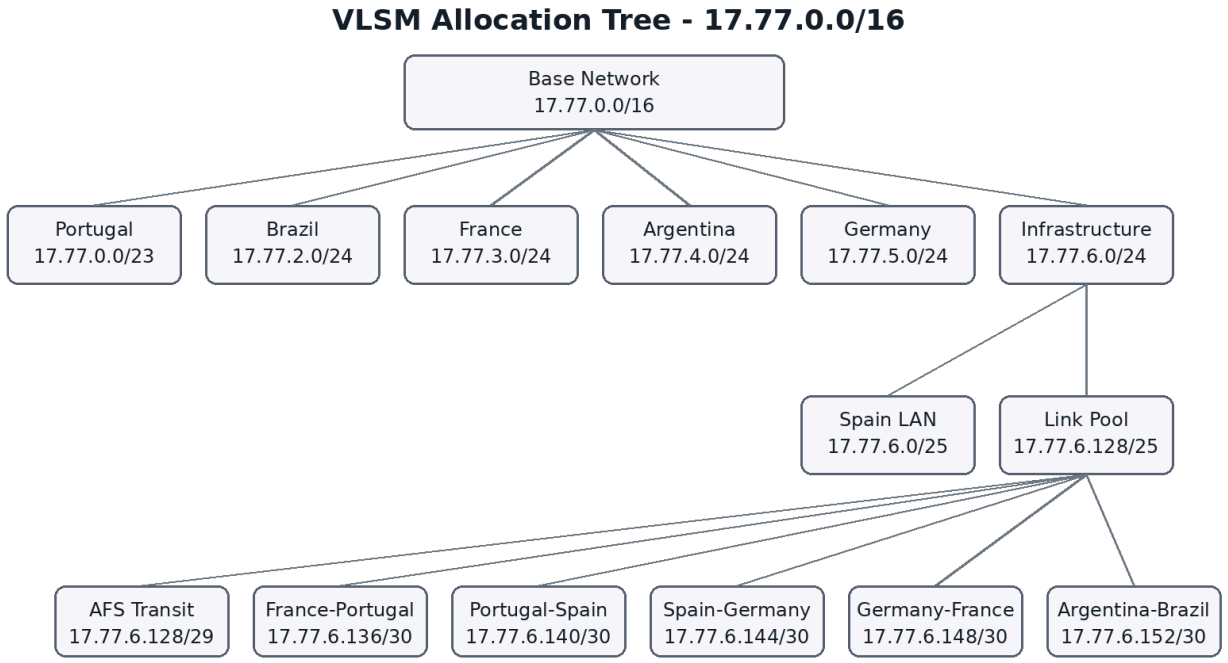


Figure 1. VLSM allocation tree derived from 17.77.0.0/16.

4.3 VLSM Subnet Table

Network	Hosts	CIDR	Mask	Usable Range	Broadcast	Capacity
Portugal LAN	300	17.77.0.0/23	255.255.254.0	17.77.0.1 - 17.77.1.254	17.77.1.255	510
Brazil LAN	250	17.77.2.0/24	255.255.255.0	17.77.2.1 - 17.77.2.254	17.77.2.255	254
France LAN	200	17.77.3.0/24	255.255.255.0	17.77.3.1 - 17.77.3.254	17.77.3.255	254
Argentina LAN	200	17.77.4.0/24	255.255.255.0	17.77.4.1 - 17.77.4.254	17.77.4.255	254
Germany LAN	150	17.77.5.0/24	255.255.255.0	17.77.5.1 - 17.77.5.254	17.77.5.255	254
Spain LAN	100	17.77.6.0/25	255.255.255.128	17.77.6.1 - 17.77.6.126	17.77.6.127	126
AFS Transit	3	17.77.6.128/29	255.255.255.248	17.77.6.129 - 17.77.6.134	17.77.6.135	6
France-Portugal	2	17.77.6.136/30	255.255.255.252	17.77.6.137 - 17.77.6.138	17.77.6.139	2
Portugal-Spain	2	17.77.6.140/30	255.255.255.252	17.77.6.141 - 17.77.6.142	17.77.6.143	2
Spain-Germany	2	17.77.6.144/30	255.255.255.252	17.77.6.145 - 17.77.6.146	17.77.6.147	2
Germany-France	2	17.77.6.148/30	255.255.255.252	17.77.6.149 - 17.77.6.150	17.77.6.151	2
Argentina-Brazil	2	17.77.6.152/30	255.255.255.252	17.77.6.153 - 17.77.6.154	17.77.6.155	2

Table 2. VLSM allocation and usable address ranges.

4.4 Router Interface Address Table

Router	Interface	IP / Prefix	Connection
France	G0/0	17.77.3.1/24	France LAN
France	G0/1	17.77.6.129/29	AFS transit
France	S0/0/0	17.77.6.137/30	Portugal
France	S0/0/1	17.77.6.150/30	Germany
Spain	G0/0	17.77.6.1/25	Spain LAN
Spain	G0/1	17.77.6.130/29	AFS transit
Spain	S0/0/0	17.77.6.142/30	Portugal
Spain	S0/0/1	17.77.6.145/30	Germany
Germany	G0/0	17.77.5.1/24	Germany LAN

Router	Interface	IP / Prefix	Connection
Germany	S0/0/0	17.77.6.146/30	Spain
Germany	S0/0/1	17.77.6.149/30	France
Portugal	G0/0	17.77.0.1/23	Portugal LAN
Portugal	S0/0/0	17.77.6.138/30	France
Portugal	S0/0/1	17.77.6.141/30	Spain
Argentina	G0/0	17.77.4.1/24	Argentina LAN
Argentina	G0/1	17.77.6.131/29	AFS transit
Argentina	S0/0/0	17.77.6.153/30	Brazil
Brazil	G0/0	17.77.2.1/24	Brazil LAN
Brazil	S0/0/0	17.77.6.154/30	Argentina

Table 3. Router interface addressing.

4.5 Server, Printer and Observed Client Addresses

Hotel	Device / Service	IP Address	Assignment
France	Mail server	17.77.3.10	Static
France	Web server	17.77.3.11	Static
France	Printer	17.77.3.20	Static
France	PC1	17.77.3.31	DHCP observed
Spain	Mail server	17.77.6.10	Static
Spain	Printer	17.77.6.20	Static
Spain	PC1	17.77.6.21	DHCP observed
Germany	Mail server	17.77.5.10	Static
Germany	Printer	17.77.5.20	Static
Germany	PC1	17.77.5.31	DHCP observed
Portugal	Mail server	17.77.0.10	Static
Portugal	Printer	17.77.0.20	Static
Portugal	PC1	17.77.0.31	DHCP observed
Argentina	DHCP server	17.77.4.10	Static
Argentina	DNS server	17.77.4.11	Static
Argentina	Mail server	17.77.4.12	Static
Argentina	Web server	17.77.4.13	Static
Argentina	Printer	17.77.4.20	Static
Brazil	DHCP server	17.77.2.10	Static
Brazil	Mail server	17.77.2.11	Static
Brazil	Printer	17.77.2.20	Static

Table 4. Static infrastructure addresses and client addresses visible in the supplied evidence.

5. Network Topology and Device Roles

The completed Packet Tracer topology implements the required six hotel routers, separate hotel LANs, the France-Spain-Argentina shared switch, the European router loop, and the exclusive Argentina-Brazil link. End devices and servers are grouped by country to make verification and troubleshooting straightforward.

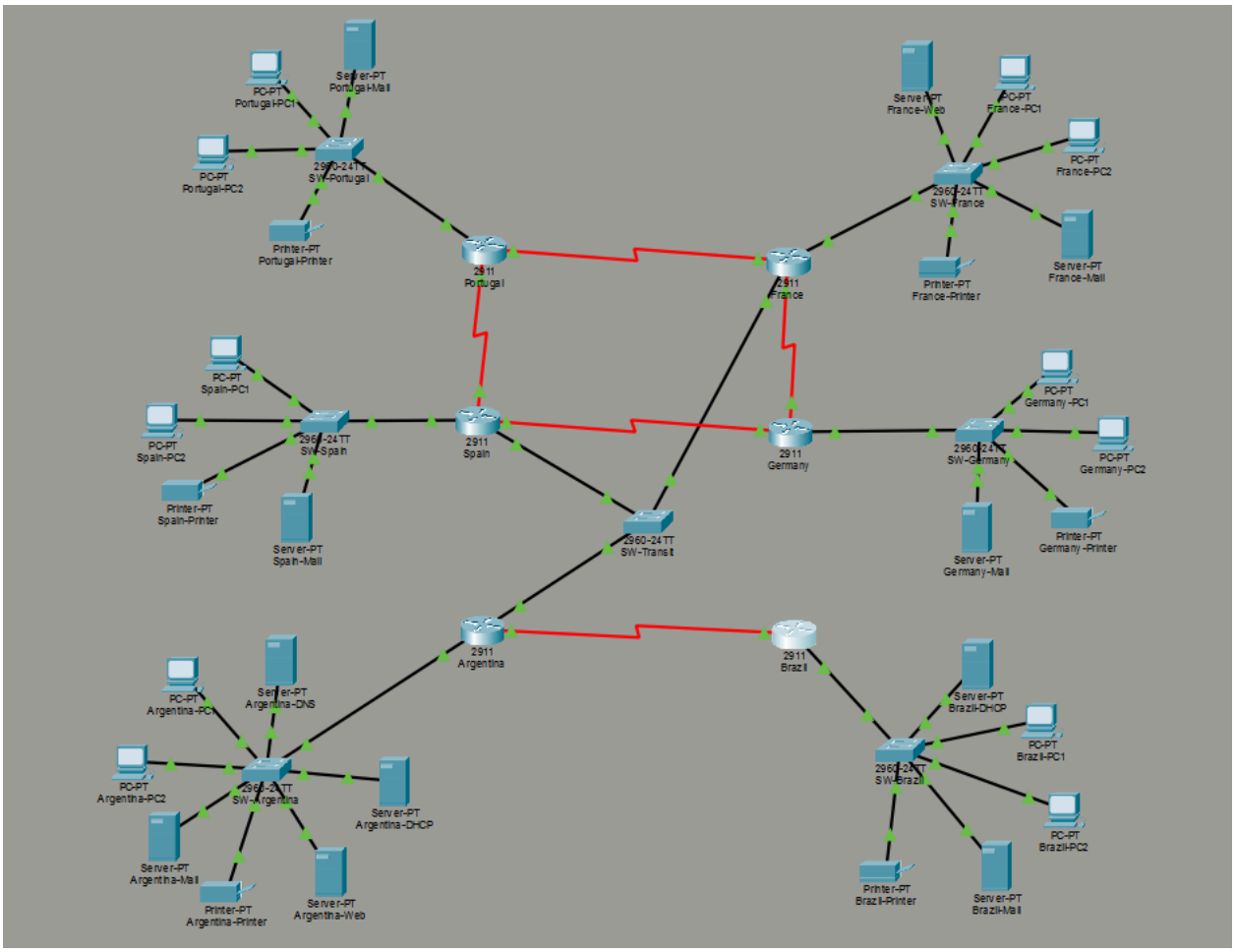


Figure 2. Completed Cisco Packet Tracer topology.

Router / Hotel	Primary Network Role
France	RIP participant; DHCP server for France/Spain/Germany/Portugal; web and mail services; static reachability toward Argentina/Brazil and Portugal.
Spain	RIP participant; shared-switch member; static path to Portugal; DHCP relay to France.
Germany	RIP participant; DHCP relay to France.
Portugal	Static routing with primary path to France and higher-distance backup path to Spain; DHCP relay to France.
Argentina	Static routing hub for Brazil; central DNS; dedicated DHCP; web and mail services.
Brazil	Default route only toward Argentina; dedicated DHCP and mail service.

Table 5. Functional role of each hotel router.

6. Network Configuration

6.1 Interface Configuration and Status

Router LAN, transit and serial interfaces were configured with the addresses listed in Table 3 and enabled with no shutdown. The France interface summary confirms that the required active interfaces are operational, while unused interfaces remain administratively down.

```
France#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     17.77.3.1       YES manual up                    up
GigabitEthernet0/1     17.77.6.129     YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Serial0/0/0            17.77.6.137     YES manual up                    up
Serial0/0/1            17.77.6.150     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
France#
```

Figure 3. France router interface status after configuration.

6.2 Routing Design

RIPv2 is used among France, Spain, and Germany. Auto-summary is disabled because the design uses multiple VLSM subnets within the same 17.77.0.0/16 major network. Static routes are used where required by the project, and Brazil uses a default route to Argentina. Portugal is configured with a primary default route through France and a floating backup route through Spain.

6.3 DHCP Configuration

France router provides DHCP to France, Spain, Germany, and Portugal. The remote hotel LAN interfaces use ip helper-address 17.77.3.1 so DHCP broadcasts can be forwarded to the France router. Argentina and Brazil use dedicated DHCP servers. Static address ranges are excluded from dynamic allocation to protect gateways, servers, and printers.

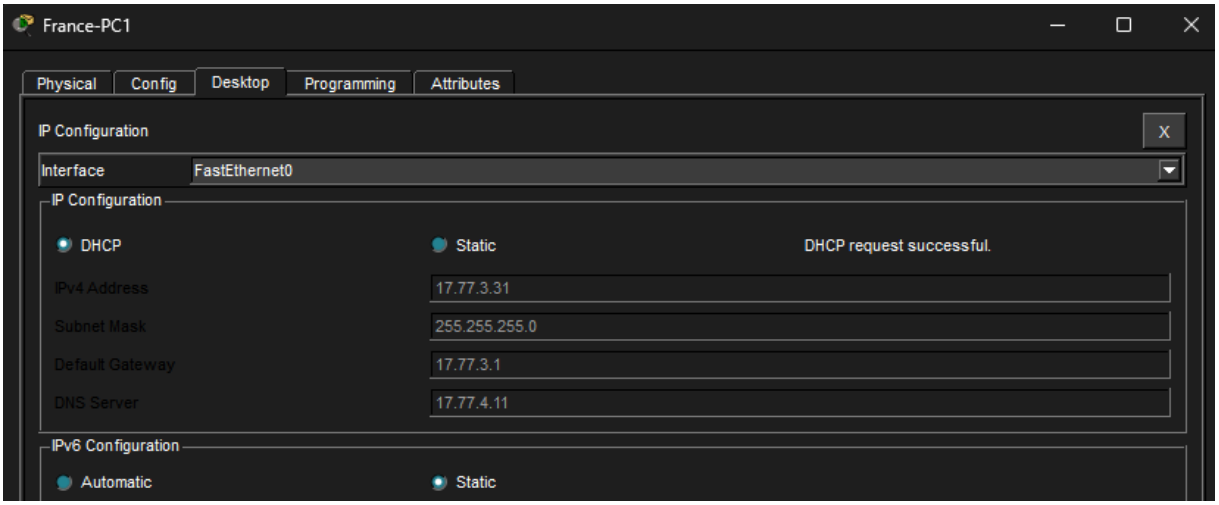


Figure 4. France-PC1 receiving 17.77.3.31/24 through DHCP.

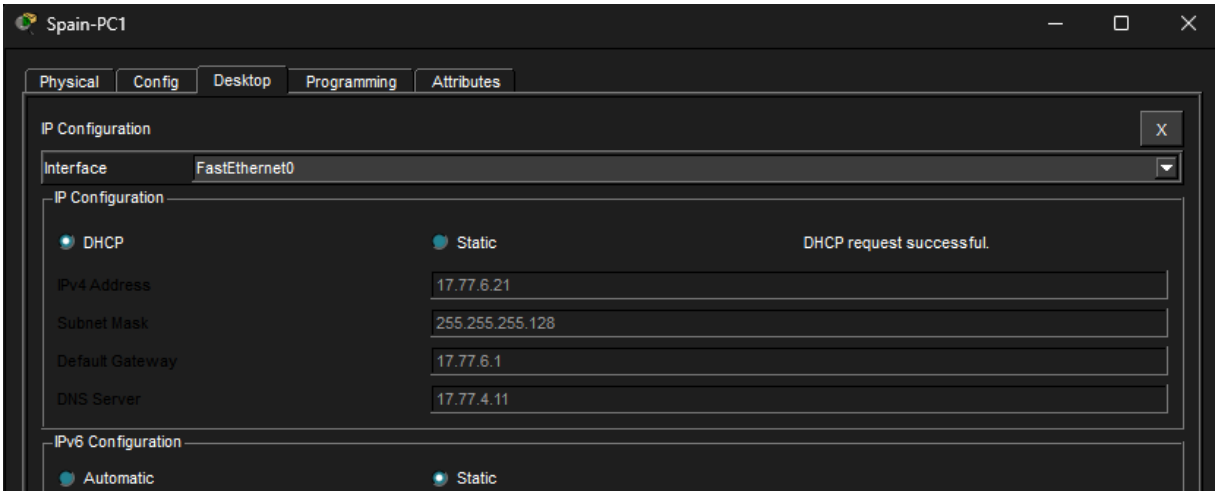


Figure 5. Spain-PC1 receiving 17.77.6.21/25 through DHCP relay.

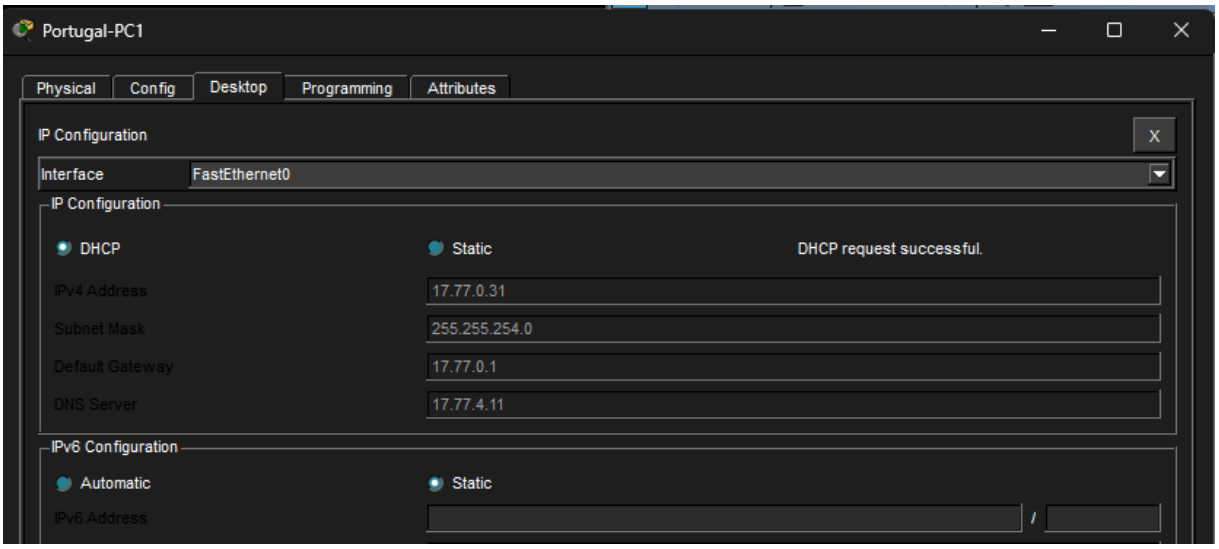


Figure 6. Portugal-PC1 receiving 17.77.0.31/23 through DHCP relay.

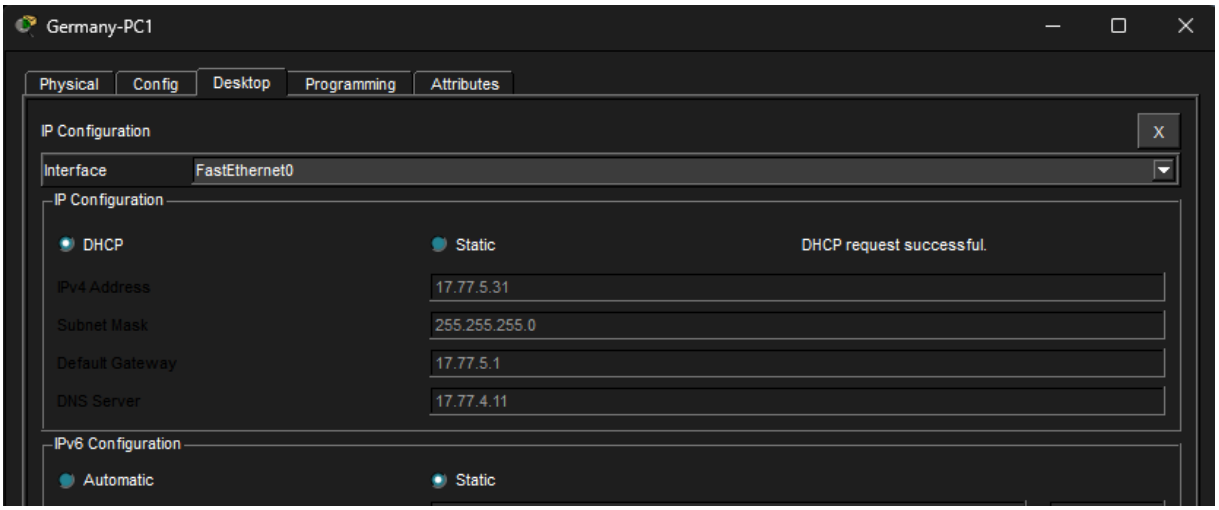


Figure 7. Germany-PC1 receiving 17.77.5.31/24 through DHCP relay.

6.4 DNS Configuration

Argentina hosts the centralized DNS server at 17.77.4.11. A records map the six mail hostnames to their respective mail servers and map the two FIFA web hostnames to the Argentina and France web servers. All DHCP pools provide 17.77.4.11 as the DNS server address.

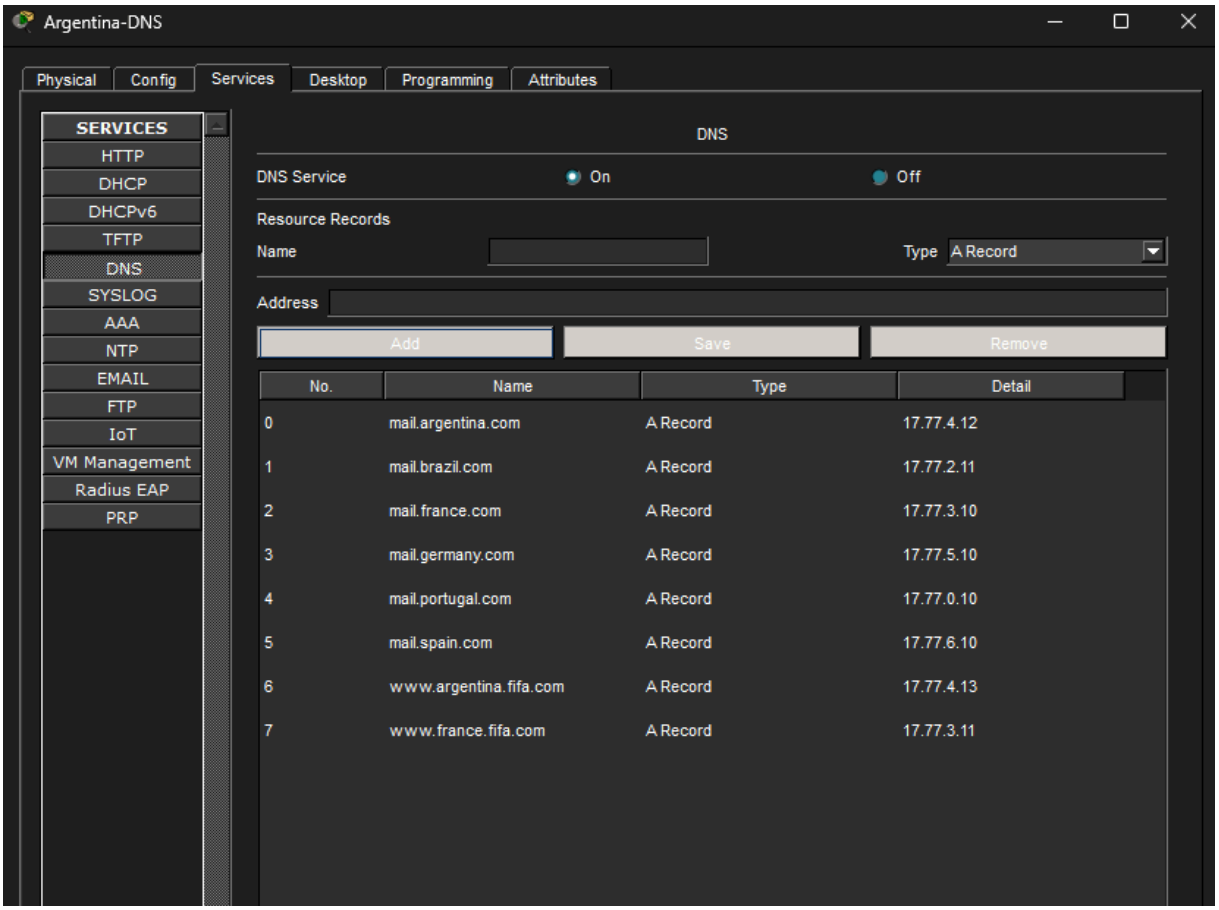


Figure 8. Central DNS service in Argentina with mail and web A records.

6.5 Web Services

HTTP is enabled on the France web server at 17.77.3.11 and the Argentina web server at 17.77.4.13. DNS records www.france.fifa.com and www.argentina.fifa.com allow users in remote hotels to access the servers by hostname rather than by IP address.

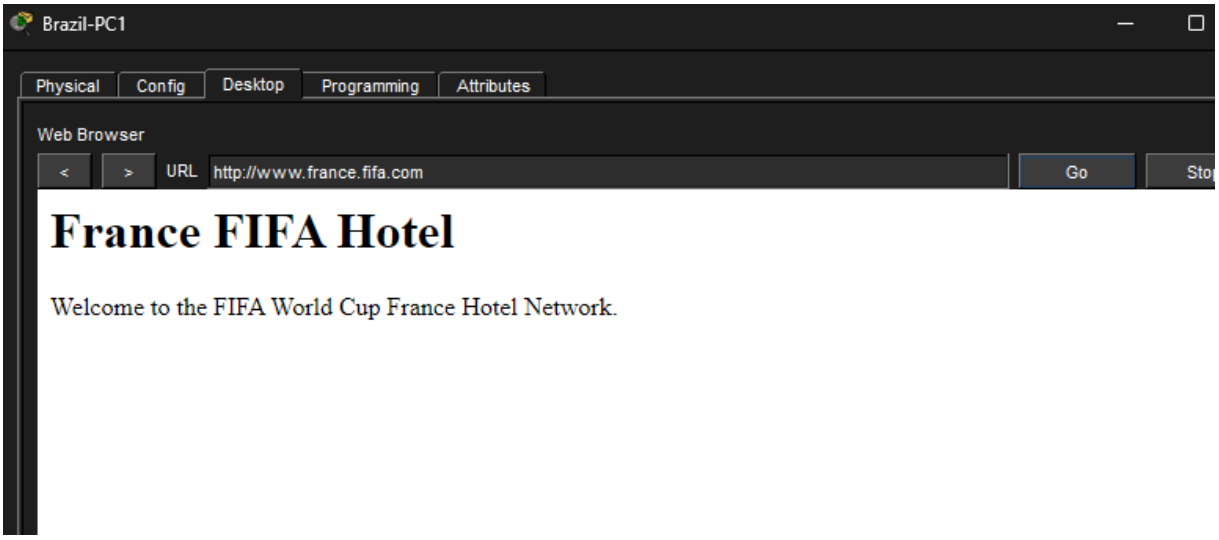


Figure 9. Brazil-PC1 successfully accessing www.france.fifa.com.

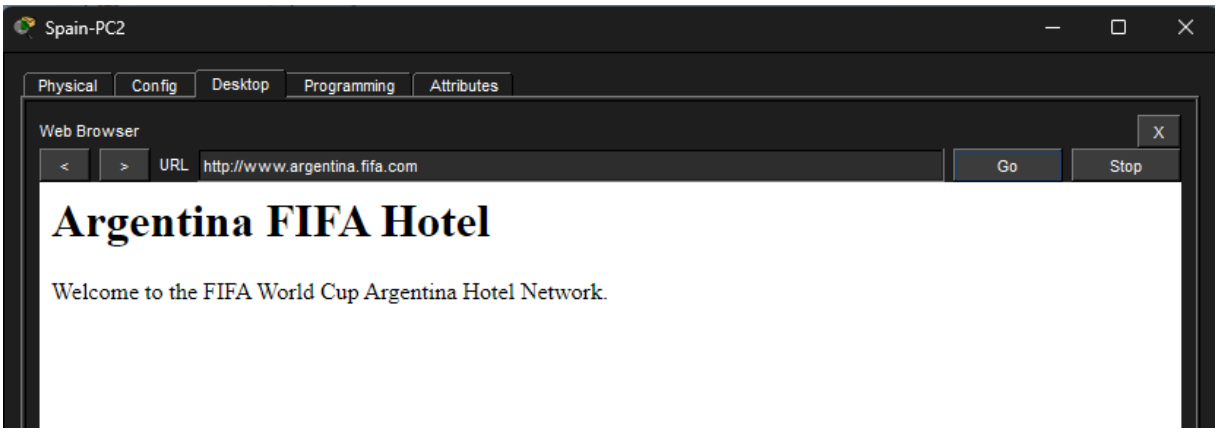


Figure 10. Spain-PC2 successfully accessing www.argentina.fifa.com.

6.6 Email Services

Each hotel has an email server with SMTP and POP3 enabled. The mail server hostnames follow the required pattern mail.<country>.com. The configured example accounts use country-specific domains, allowing cross-hotel email communication. The supplied test evidence shows a reply from Argentina successfully received by the France user.

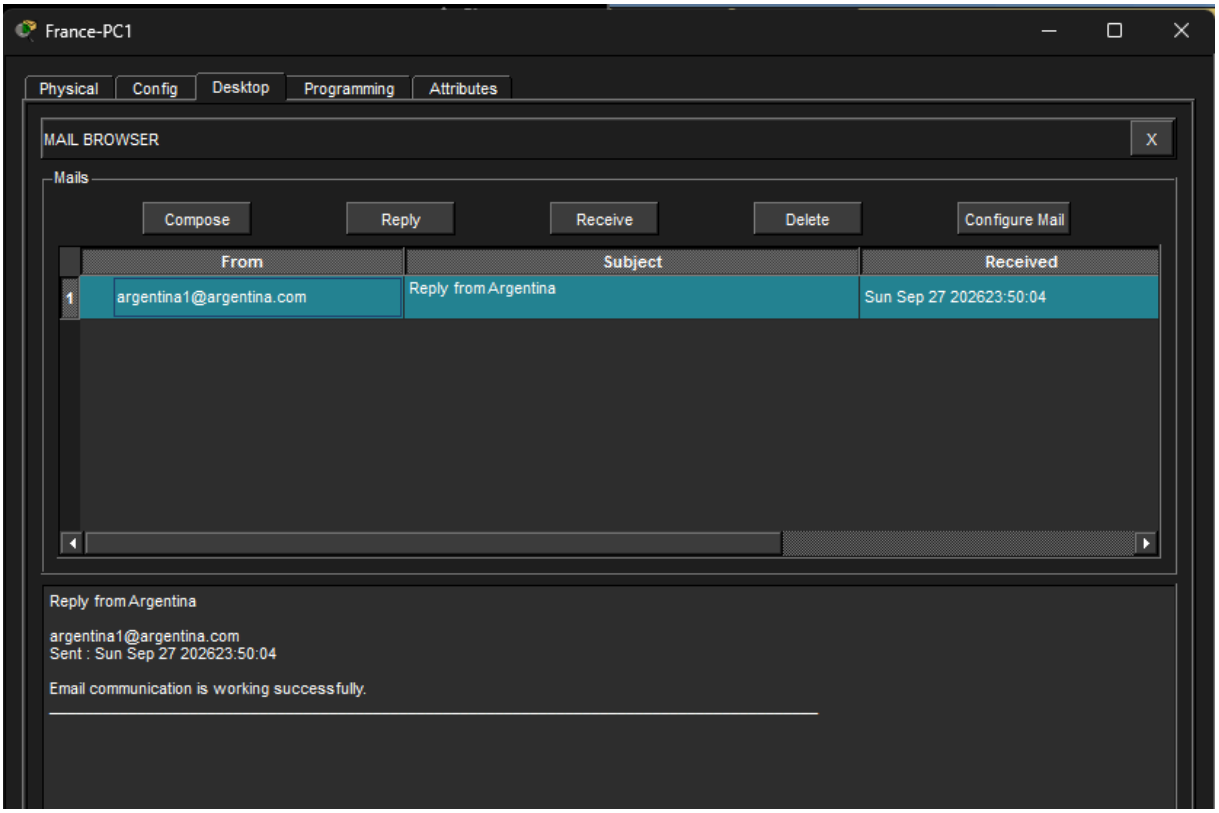


Figure 11. Successful inter-hotel email communication between Argentina and France.

7. Verification and Test Results

7.1 RIP Route Learning

Germany and Spain display routes marked with R, confirming that RIPv2 is exchanging routing information among the European RIP routers. The learned routes include hotel LANs and infrastructure subnets redistributed through the participating routers.

```
Germany#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     17.0.0.0/8 is variably subnetted, 14 subnets, 6 masks
R       17.77.0.0/23 [120/1] via 17.77.6.150, 00:00:14, Serial0/0/1
R       17.77.2.0/24 [120/1] via 17.77.6.150, 00:00:14, Serial0/0/1
R       17.77.3.0/24 [120/1] via 17.77.6.150, 00:00:14, Serial0/0/1
R       17.77.4.0/24 [120/1] via 17.77.6.150, 00:00:14, Serial0/0/1
C       17.77.5.0/24 is directly connected, GigabitEthernet0/0
L       17.77.5.1/32 is directly connected, GigabitEthernet0/0
R       17.77.6.0/25 [120/1] via 17.77.6.145, 00:00:14, Serial0/0/0
R       17.77.6.128/29 [120/1] via 17.77.6.145, 00:00:14, Serial0/0/0
                       [120/1] via 17.77.6.150, 00:00:14, Serial0/0/1
R       17.77.6.136/30 [120/1] via 17.77.6.150, 00:00:14, Serial0/0/1
R       17.77.6.140/30 [120/1] via 17.77.6.145, 00:00:14, Serial0/0/0
C       17.77.6.144/30 is directly connected, Serial0/0/0
L       17.77.6.146/32 is directly connected, Serial0/0/0
C       17.77.6.148/30 is directly connected, Serial0/0/1
L       17.77.6.149/32 is directly connected, Serial0/0/1
```

Figure 12. Germany routing table showing RIP-learned routes.

```
Spain#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     17.0.0.0/8 is variably subnetted, 15 subnets, 6 masks
S       17.77.0.0/23 [1/0] via 17.77.6.141
R       17.77.2.0/24 [120/2] via 17.77.6.146, 00:00:18, Serial0/0/1
R       17.77.3.0/24 [120/1] via 17.77.6.129, 00:00:28, GigabitEthernet0/1
R       17.77.4.0/24 [120/2] via 17.77.6.146, 00:00:18, Serial0/0/1
R       17.77.5.0/24 [120/1] via 17.77.6.146, 00:00:18, Serial0/0/1
C       17.77.6.0/25 is directly connected, GigabitEthernet0/0
L       17.77.6.1/32 is directly connected, GigabitEthernet0/0
C       17.77.6.128/29 is directly connected, GigabitEthernet0/1
L       17.77.6.130/32 is directly connected, GigabitEthernet0/1
R       17.77.6.136/30 [120/1] via 17.77.6.129, 00:00:28, GigabitEthernet0/1
C       17.77.6.140/30 is directly connected, Serial0/0/0
L       17.77.6.142/32 is directly connected, Serial0/0/0
C       17.77.6.144/30 is directly connected, Serial0/0/1
L       17.77.6.145/32 is directly connected, Serial0/0/1
R       17.77.6.148/30 [120/1] via 17.77.6.129, 00:00:28, GigabitEthernet0/1
                       [120/1] via 17.77.6.146, 00:00:18, Serial0/0/1
```

Figure 13. Spain routing table showing RIP-learned routes.

7.2 Brazil Default Route

Brazil must reach every non-local network through Argentina. The routing table shows a candidate default route, S*, via 17.77.6.153, which is Argentina's serial address on the 17.77.6.152/30 link.

```
Brazil#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 17.77.6.153 to network 0.0.0.0

     17.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C       17.77.2.0/24 is directly connected, GigabitEthernet0/0
L       17.77.2.1/32 is directly connected, GigabitEthernet0/0
C       17.77.6.152/30 is directly connected, Serial0/0/0
L       17.77.6.154/32 is directly connected, Serial0/0/0
S*   0.0.0.0/0 [1/0] via 17.77.6.153

Brazil#
```

Figure 14. Brazil routing table showing the static default route via Argentina.

7.3 End-to-End Connectivity

Cross-country pings were performed from Brazil toward European networks. The captured test shows successful replies to the France gateway, the Spain gateway, and a France client address. The first probe to one destination times out before subsequent replies, which is consistent with initial address-resolution behavior in Packet Tracer; the remaining responses confirm reachability.

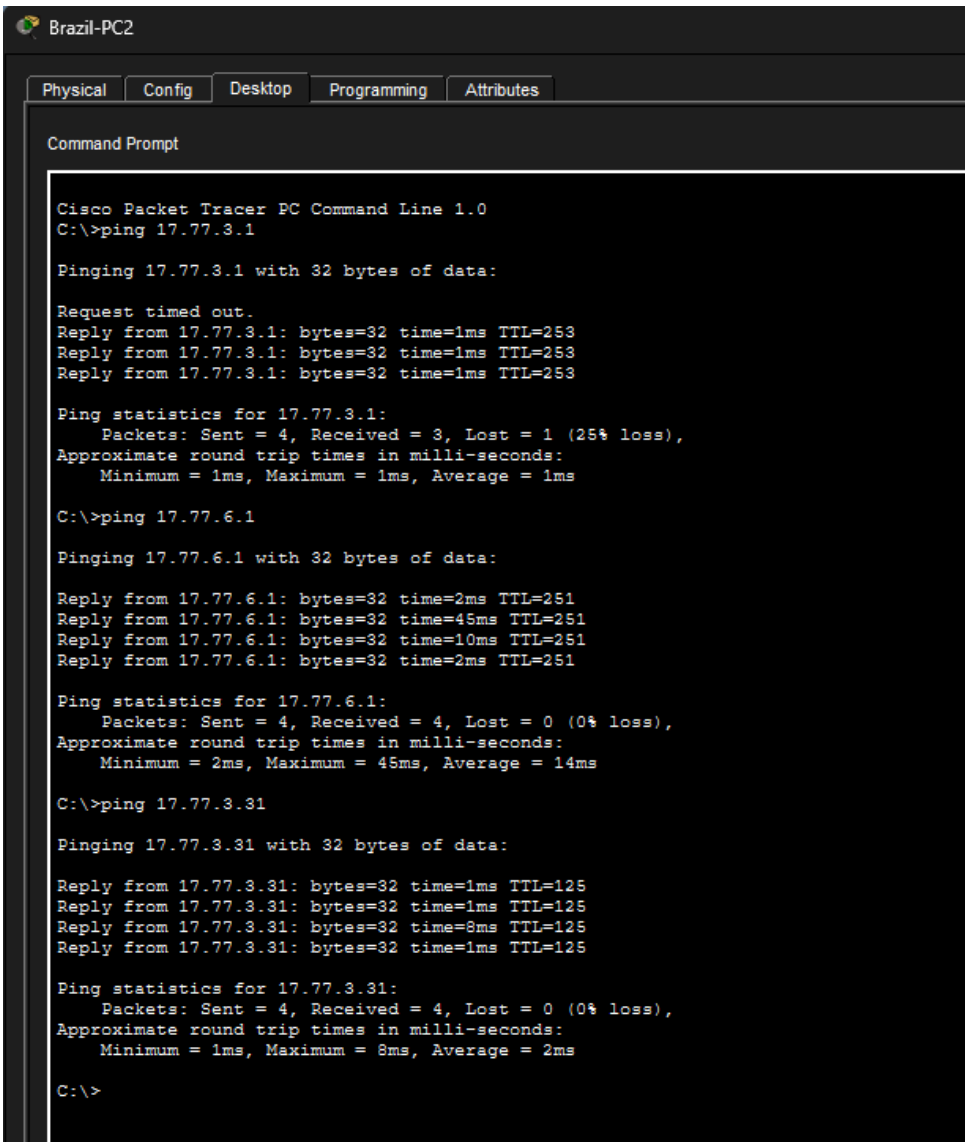


Figure 15. Successful end-to-end ICMP tests from Brazil to European destinations.

7.4 Web, DNS and Email Validation

The DNS screenshot confirms that the required mail and web hostnames are present. Browser screenshots show that remote clients can reach both FIFA websites by hostname, demonstrating combined routing, DNS and HTTP functionality. The email screenshot confirms application-layer communication between two hotel domains.

7.5 Link-Failure / Backup-Path Test

The direct Portugal-France serial path was deliberately disabled to test resilience. Portugal is configured with a primary default route through France and a higher-administrative-distance backup route through Spain. The supplied topology excerpt shows the link-under-test, and the CLI evidence records the France serial interface shutdown followed by continued successful reachability to 17.77.4.1.

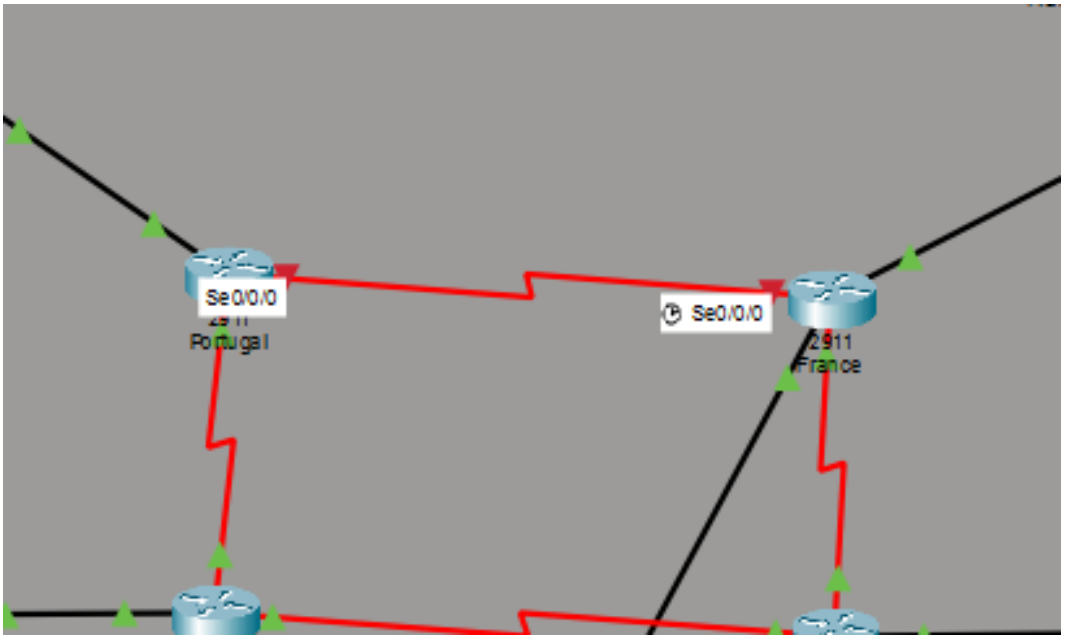


Figure 16. European loop segment used for the Portugal-France link-failure test.



Figure 17. Link-down test evidence: France S0/0/0 shut down and successful reachability maintained.

Evidence note: the source screenshot document labels the final image as Portugal's show ip route, but the CLI prompt visible in the supplied image is France#. Therefore, this report does not claim that the screenshot itself proves Portugal's selected next-hop; the backup path is documented by the Portugal router configuration in Appendix A.

7.6 Verification Matrix

Test	Evidence	Result
Physical/interface status	France show ip interface brief	Pass - required configured interfaces are up/up.
RIP route exchange	Germany and Spain show ip route	Pass - multiple R routes are present.
Brazil default routing	Brazil show ip route	Pass - S* default route via 17.77.6.153.
DHCP - France	France-PC1 IP configuration	Pass - 17.77.3.31/24, gateway 17.77.3.1, DNS 17.77.4.11.
DHCP - Spain	Spain-PC1 IP configuration	Pass - 17.77.6.21/25, gateway 17.77.6.1.
DHCP - Portugal	Portugal-PC1 IP configuration	Pass - 17.77.0.31/23, gateway 17.77.0.1.
DHCP - Germany	Germany-PC1 IP configuration	Pass - 17.77.5.31/24, gateway 17.77.5.1.
Central DNS	Argentina DNS service	Pass - required mail and web A records present.
France web service	Brazil-PC1 browser	Pass - www.france.fifa.com loads successfully.
Argentina web service	Spain-PC2 browser	Pass - www.argentina.fifa.com loads successfully.
Email	France-PC1 mail client	Pass - reply from Argentina received.
End-to-end ping	Brazil-PC2 command prompt	Pass - France and Spain destinations reachable.
Link-failure test	France serial shutdown + continued ping	Pass for continued reachability; screenshot does not directly show Portugal routing table.

Table 6. Verification summary.

8. Work Distribution

The supplied submission information identifies Student ID 22101777. In the absence of additional group-member details, the report records the implementation as an individual contribution. If this project was completed by a group, this table should be replaced with the actual member allocation before submission.

Student ID	Work Performed
22101777	VLSM design, topology construction, router addressing, RIP/static routing, DHCP, DNS, web and email service configuration, connectivity testing, failover testing, screenshots and report preparation.

Table 7. Work distribution recorded for the supplied submission.

9. Conclusion

The FIFA World Cup hotel network was designed and implemented in Cisco Packet Tracer using the student-ID-derived address block 17.77.0.0/16. VLSM provides appropriate subnet sizes for all six hotels and for the inter-router links. The final topology combines RIPv2, static routes, a Brazil default route, DHCP, centralized DNS, HTTP, SMTP/POP3 email, and a redundant European path.

The supplied verification screenshots demonstrate the core functional requirements: interfaces are operational, RIP routes are learned, Brazil forwards through Argentina, client addressing is dynamic where required, both web services resolve and load from remote hotels, email communication succeeds between hotels, and cross-country pings complete successfully. The link-failure test also confirms that the network remains operational after the direct France-Portugal serial interface is shut down.

Appendix A. Router Configuration Commands

Interface identifiers may vary slightly depending on the Packet Tracer HWIC slot used. The commands below reflect the addressing and routing design implemented in this report.

France

```
enable
configure terminal
hostname France
interface g0/0
 ip address 17.77.3.1 255.255.255.0
 no shutdown
exit
interface g0/1
 ip address 17.77.6.129 255.255.255.248
 no shutdown
exit
interface s0/0/0
 ip address 17.77.6.137 255.255.255.252
 clock rate 64000
 no shutdown
exit
interface s0/0/1
 ip address 17.77.6.150 255.255.255.252
 no shutdown
exit
ip route 17.77.4.0 255.255.255.0 17.77.6.131
ip route 17.77.2.0 255.255.255.0 17.77.6.131
ip route 17.77.0.0 255.255.254.0 17.77.6.138
router rip
 version 2
 no auto-summary
 network 17.0.0.0
 redistribute static
exit
ip dhcp excluded-address 17.77.3.1 17.77.3.30
```

```
ip dhcp excluded-address 17.77.5.1 17.77.5.30
ip dhcp excluded-address 17.77.6.1 17.77.6.20
ip dhcp excluded-address 17.77.0.1 17.77.0.30
ip dhcp pool FRANCE
 network 17.77.3.0 255.255.255.0
 default-router 17.77.3.1
 dns-server 17.77.4.11
exit
ip dhcp pool GERMANY
 network 17.77.5.0 255.255.255.0
 default-router 17.77.5.1
 dns-server 17.77.4.11
exit
ip dhcp pool SPAIN
 network 17.77.6.0 255.255.255.128
 default-router 17.77.6.1
 dns-server 17.77.4.11
exit
ip dhcp pool PORTUGAL
 network 17.77.0.0 255.255.254.0
 default-router 17.77.0.1
 dns-server 17.77.4.11
exit
end
write memory
```

Spain

```
enable
configure terminal
hostname Spain
interface g0/0
 ip address 17.77.6.1 255.255.255.128
 ip helper-address 17.77.3.1
 no shutdown
exit
interface g0/1
 ip address 17.77.6.130 255.255.255.248
 no shutdown
exit
interface s0/0/0
 ip address 17.77.6.142 255.255.255.252
 no shutdown
exit
interface s0/0/1
 ip address 17.77.6.145 255.255.255.252
 clock rate 64000
 no shutdown
exit
ip route 17.77.0.0 255.255.254.0 17.77.6.141
router rip
 version 2
 no auto-summary
 network 17.0.0.0
 redistribute static
exit
end
write memory
```

Germany

```
enable
configure terminal
hostname Germany
interface g0/0
 ip address 17.77.5.1 255.255.255.0
```

```
 ip helper-address 17.77.3.1
 no shutdown
exit
interface s0/0/0
 ip address 17.77.6.146 255.255.255.252
 no shutdown
exit
interface s0/0/1
 ip address 17.77.6.149 255.255.255.252
 clock rate 64000
 no shutdown
exit
router rip
 version 2
 no auto-summary
 network 17.0.0.0
exit
end
write memory
```

Portugal

```
enable
configure terminal
hostname Portugal
interface g0/0
 ip address 17.77.0.1 255.255.254.0
 ip helper-address 17.77.3.1
 no shutdown
exit
interface s0/0/0
 ip address 17.77.6.138 255.255.255.252
 no shutdown
exit
interface s0/0/1
 ip address 17.77.6.141 255.255.255.252
 clock rate 64000
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 17.77.6.137
ip route 0.0.0.0 0.0.0.0 17.77.6.142 5
end
write memory
```

Argentina

```
enable
configure terminal
hostname Argentina
interface g0/0
 ip address 17.77.4.1 255.255.255.0
 no shutdown
exit
interface g0/1
 ip address 17.77.6.131 255.255.255.248
 no shutdown
exit
interface s0/0/0
 ip address 17.77.6.153 255.255.255.252
 clock rate 64000
 no shutdown
exit
ip route 17.77.0.0 255.255.254.0 17.77.6.129
ip route 17.77.3.0 255.255.255.0 17.77.6.129
ip route 17.77.5.0 255.255.255.0 17.77.6.129
ip route 17.77.6.0 255.255.255.128 17.77.6.130
```

```
ip route 17.77.2.0 255.255.255.0 17.77.6.154
end
write memory
```

Brazil

```
enable
configure terminal
hostname Brazil
interface g0/0
 ip address 17.77.2.1 255.255.255.0
 no shutdown
exit
interface s0/0/0
 ip address 17.77.6.154 255.255.255.252
 no shutdown
exit
ip route 0.0.0.0 0.0.0.0 17.77.6.153
end
write memory
```

Appendix B. DNS and Service Records

DNS Name	Address	Service
mail.argentina.com	17.77.4.12	Argentina mail
mail.brazil.com	17.77.2.11	Brazil mail
mail.france.com	17.77.3.10	France mail
mail.germany.com	17.77.5.10	Germany mail
mail.portugal.com	17.77.0.10	Portugal mail
mail.spain.com	17.77.6.10	Spain mail
www.argentina.fifa.com	17.77.4.13	Argentina web
www.france.fifa.com	17.77.3.11	France web

Table 8. Central DNS A records shown in the supplied implementation.

Mail Domains and Test Accounts

Domain	Example User	Services
france.com	france1	SMTP + POP3
spain.com	spain1	SMTP + POP3
germany.com	germany1	SMTP + POP3
portugal.com	portugal1	SMTP + POP3
argentina.com	argentina1	SMTP + POP3
brazil.com	brazil1	SMTP + POP3

Table 9. Email domains used for inter-hotel communication.